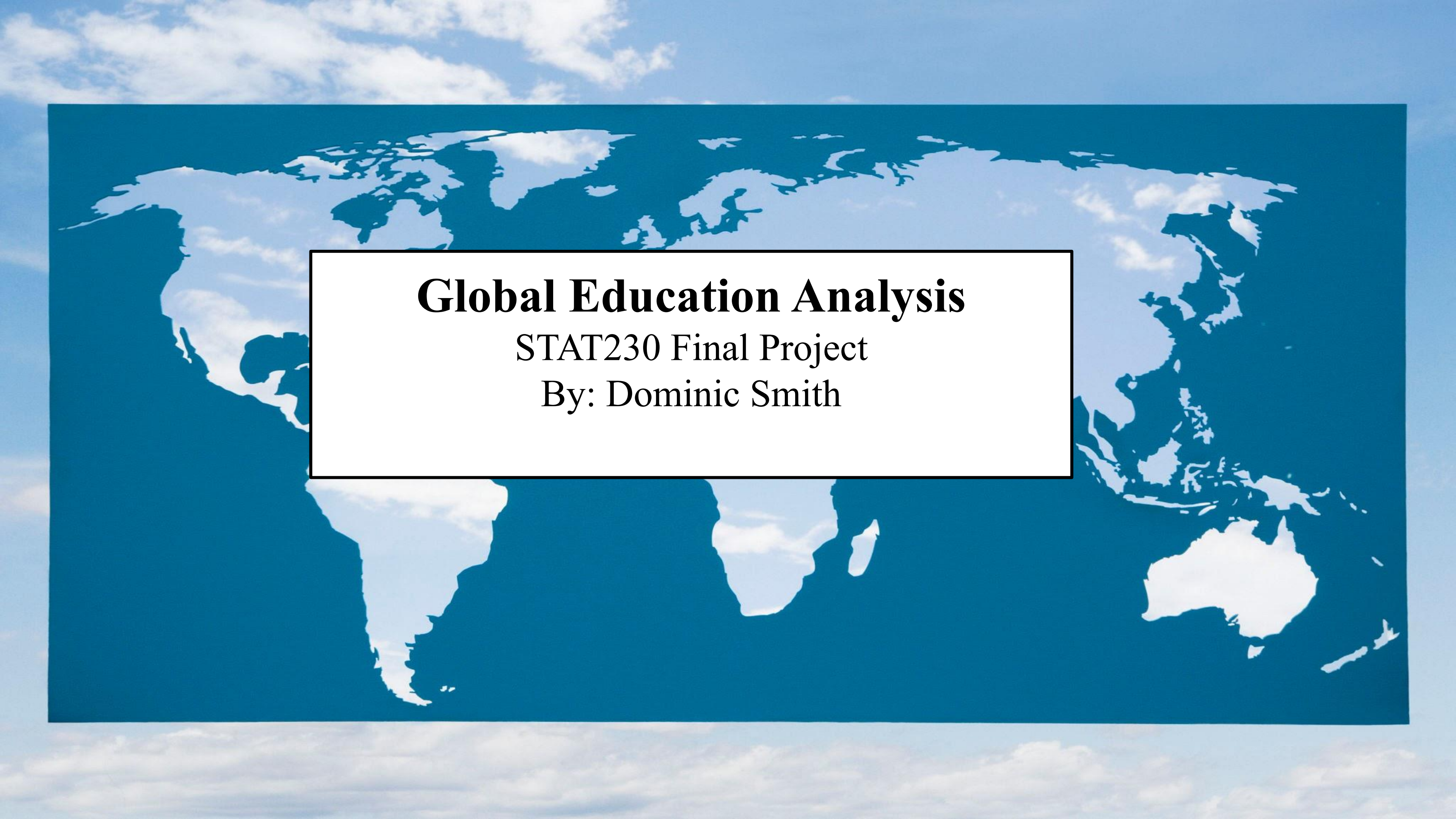

Global Education Analysis

STAT230 Final Project

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Data Variables

Country: The name of the country

Income_Group: The World Bank classification of countries *(High income, Upper middle income, Lower middle income, Low income)*

Primary_Completion_Rate: Percentage of students completing primary education

Education_Expenditure: Government expenditure on education as a percentage of GDP

Pupil_Teacher_Ratio: The average number of pupils per teacher in primary education

GDP_Per_Capita: Gross Domestic Product per capita in current US dollars

Research Questions

1.) Do richer countries have better primary school completion rates than poorer ones?

2.) Does spending more money on education lead to fewer students per teacher?

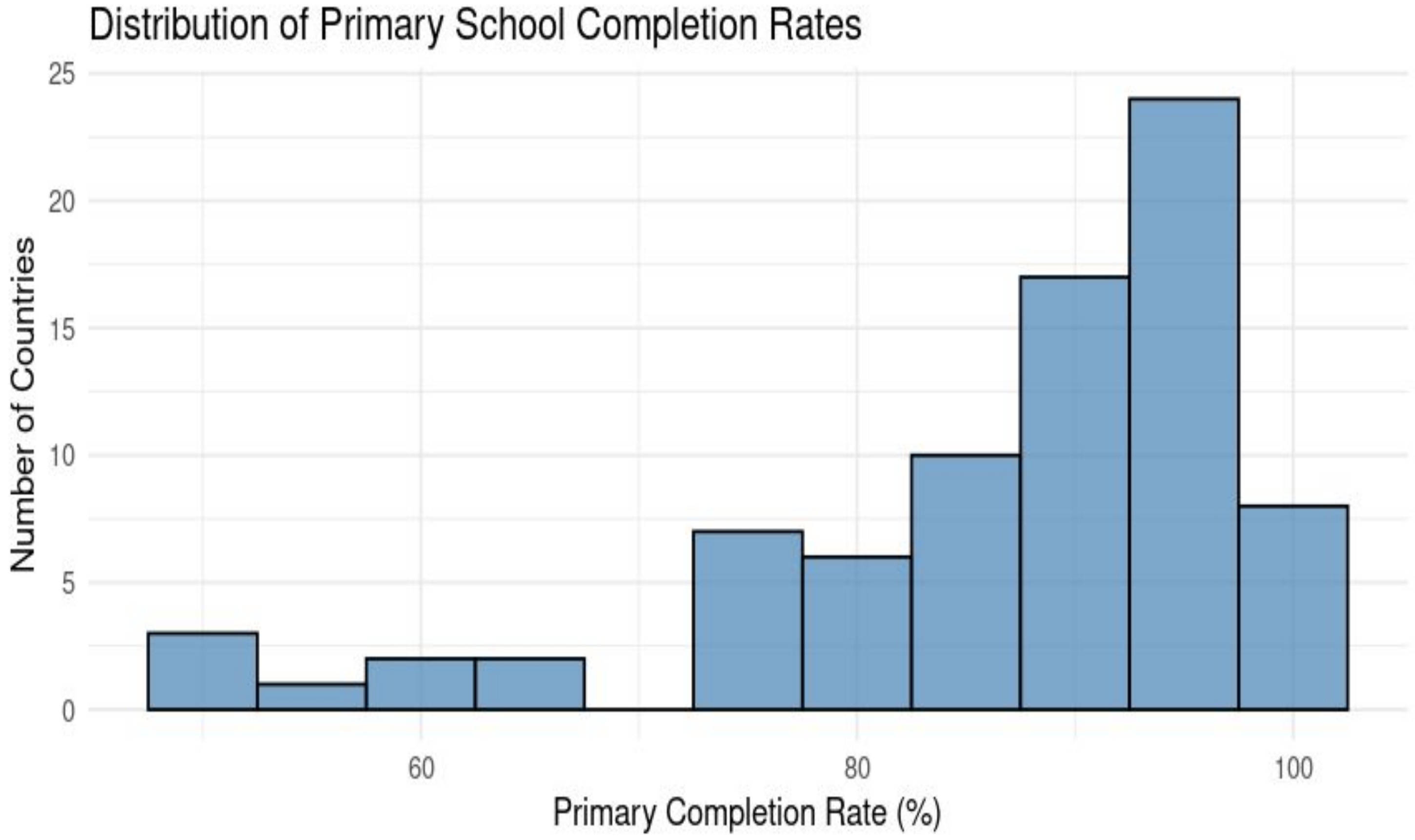
World Bank Education Data

Dataset Description

Variable	*Description*
Country	80 nations across all continents
Income Group	High income (20 countries) Upper middle (28) Lower middle (20) Low income (12)
Primary Compl. %	Range: 50% - 100% Avg: 84.7%
Education Exp.	Range: 1.5% - 7% of GDP Avg: 4.3% of GDP
Pupil-Teacher Ratio	Range: 10 - 50 students Avg: 24.5 students
GDP Per Capita	Range: $300 - $90,000 Varied by income group

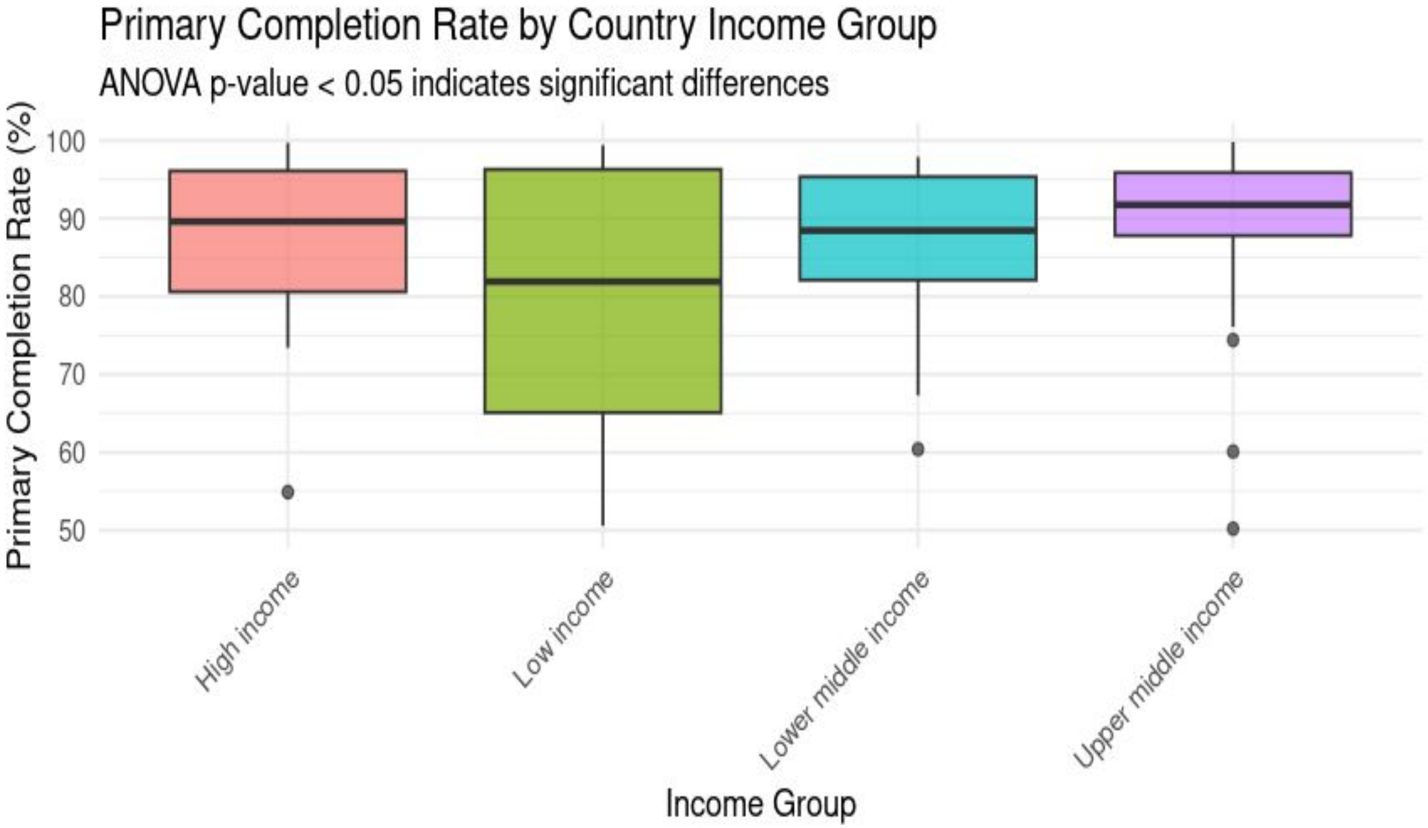
Website- https://data.worldbank.org/topic/education

Primary Completion Rate Histogram



Most countries achieve high completion rates,but notable disparities exist

Completion Rates by Income Group



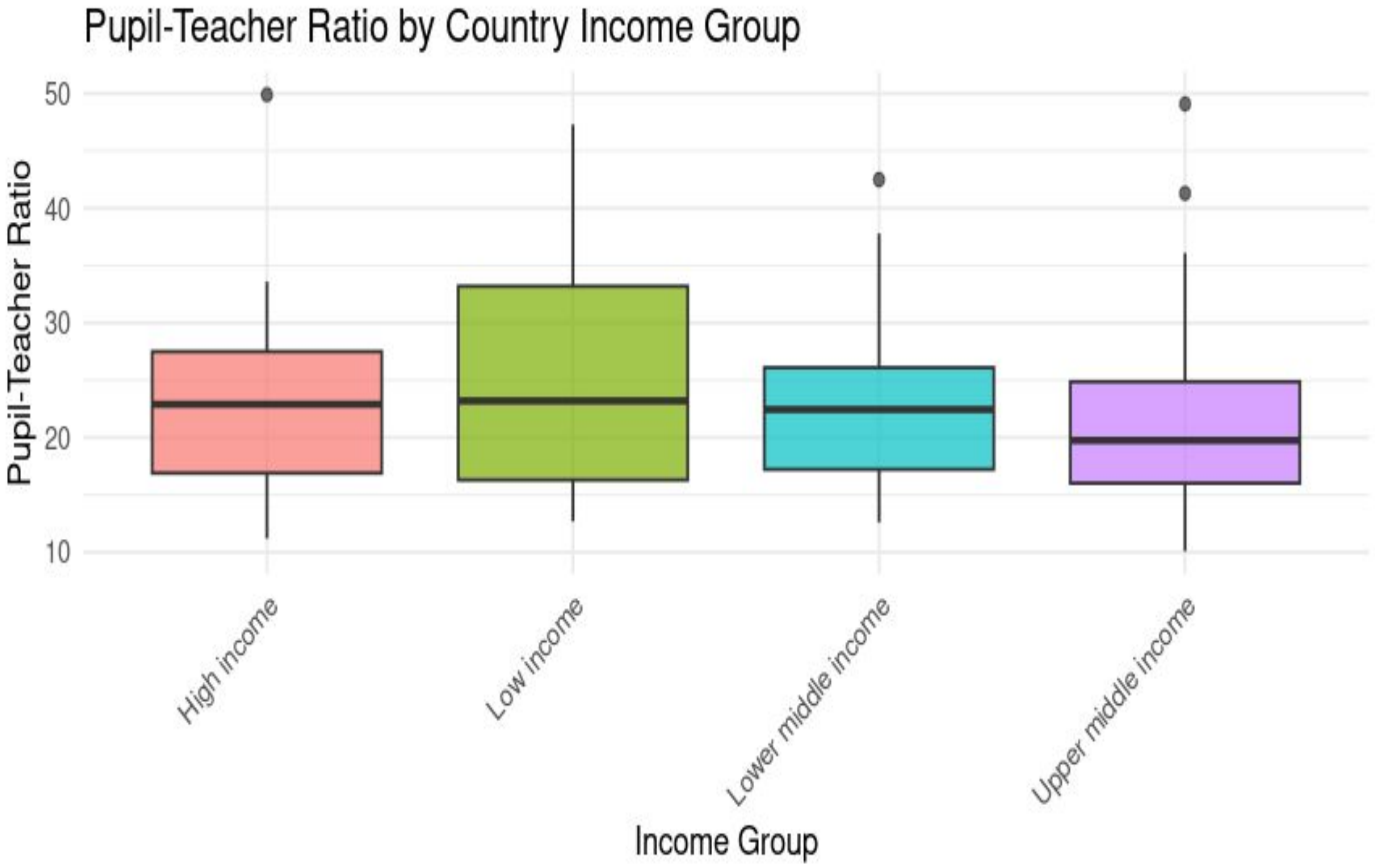
Clear pattern of decreasing completion rates with lower income levels

ANOVA Results

- **Findings:** Strong evidence of differences between income groups
- **Effects:** Income group classification explains a substantial portion of variation in primary completion rates

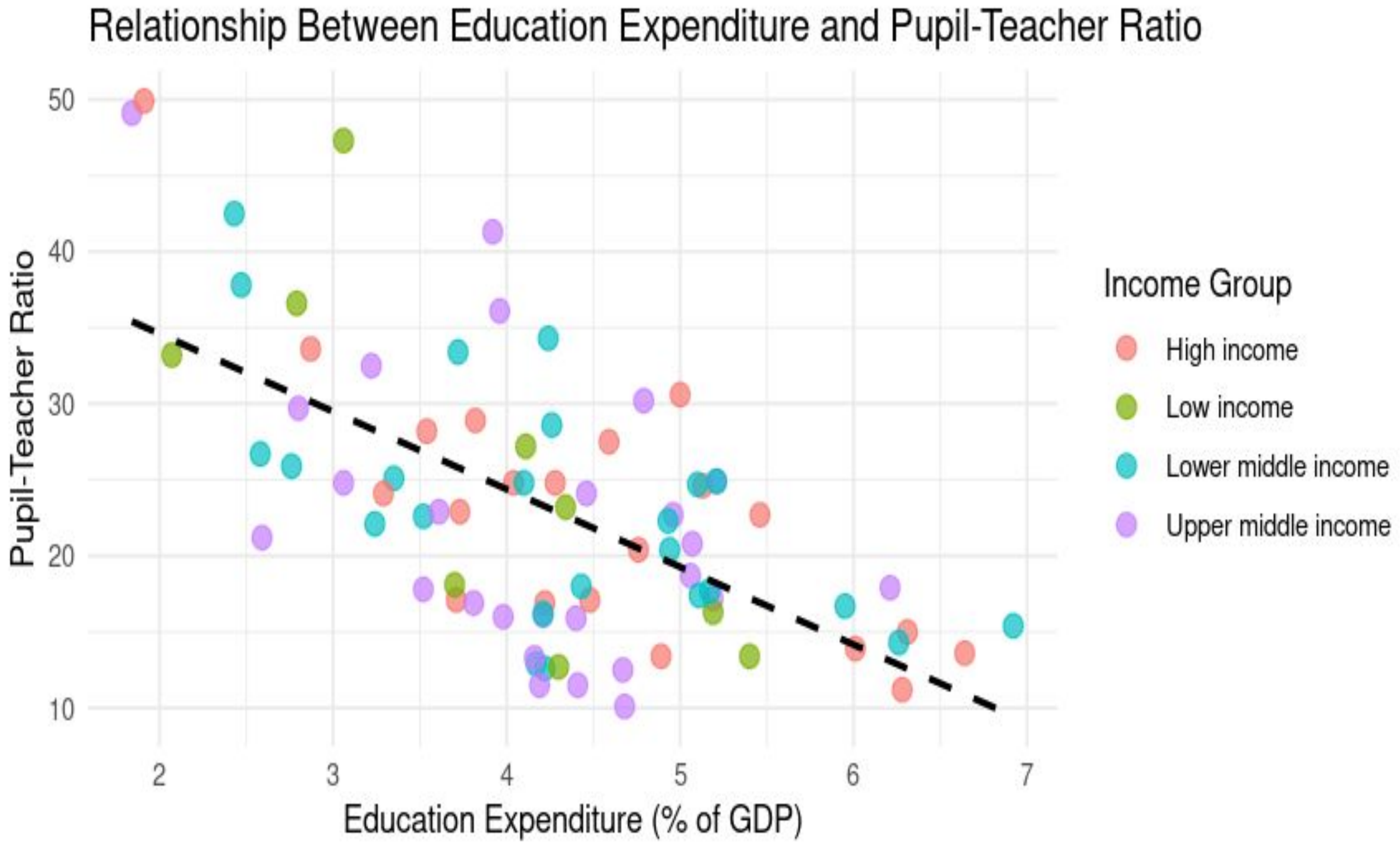
Comparison	*Difference*	*Significance*
Upper middle - High	Lower by ~5-8%	Significant
Lower middle - High	Lower by ~15-20%	Significant
Low - High	Lower by ~25-30%	Significant
Lower middle - Upper middle	Lower by ~10-12%	Significant
Low - Upper middle	Lower by ~17-22%	Significant
Low - Lower middle	Lower by ~7-10%	Significant

Pupil-Teacher Ratio by Income Group



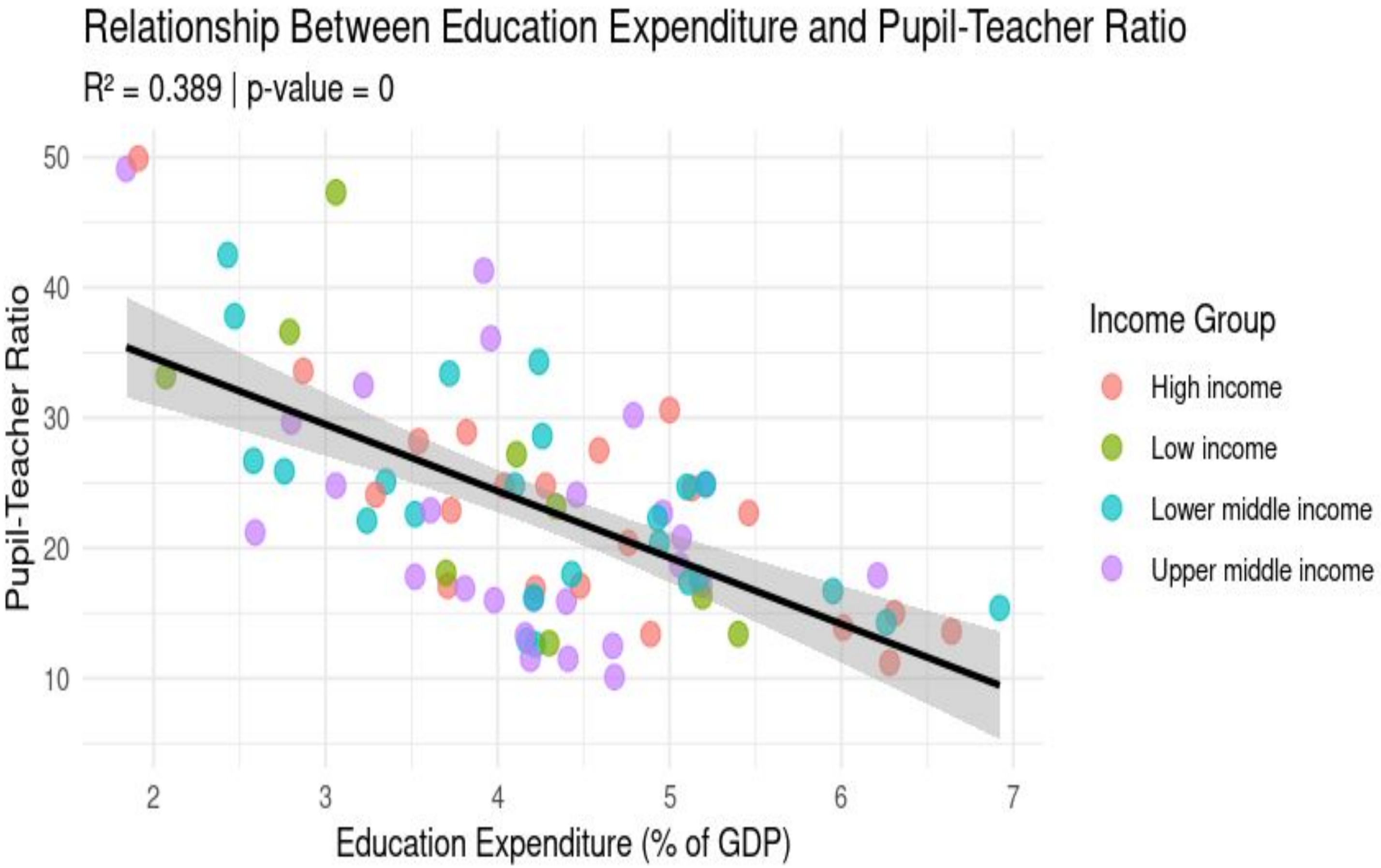
Higher-income countries maintain lower pupil-teacher ratios

Education Expenditure vs. Pupil-Teacher Ratio



Higher education expenditure correlates with lower pupil-teacher ratios

Regression Results



Interpretation

- For each additional 1% of GDP spent on education, pupil-teacher ratio decreases by approx. 3.5 students
- Education spending explains the variation in pupil-teacher ratios

Key Findings

- Richer countries have much higher primary school completion rates
- More education spending leads to smaller class sizes
- Education quality follows economic development patterns

Conclusion

In this analysis, I explored global education data focusing on primary completion rates and classroom resources across different income groups. The descriptive statistics showed clear patterns in the data, with higher-income countries generally having better educational outcomes.

For my first research question, the ANOVA results showed significant differences in primary completion rates across income groups. High-income countries had significantly higher completion rates compared to lower-income countries, demonstrating educational inequality that correlates with economic development. The Tukey's HSD test further confirmed significant differences between all income group pairs.

For my second question, I found a significant negative correlation between education expenditure and pupil-teacher ratios. The linear regression model showed that as countries increase their education expenditure (as a percentage of GDP), the pupil-teacher ratio tends to decrease, indicating improved classroom resources and potentially better learning environments. The R-squared value from the regression model suggests that education expenditure explains a substantial portion of the variation in pupil-teacher ratios across countries.

These findings highlight the importance of economic development and education investment in improving educational outcomes globally. Policymakers should consider targeted investments in education, especially in lower-income countries, to reduce educational inequality and improve classroom conditions.